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UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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*Dedicated to my family who endured missed birthdays,
missed festivals and countless days alone so that I may
make history*

Chapter 1

Introduction

The Standard Model (SM) of the particle physics gives successful explanation about the constituents of the physical matter and the fundamental forces through which they interact with each other and with composite particles. The electromagnetic, strong and weak forces are described by dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavordynamics (QFD) which are encoded in the SM. Later on, electromagnetic and weak forces among these, were unified as electroweak (EWK) force at the order of 100 GeV unification energy scale. Sheldon Glashow, Abdus Salam, and Steven Weinberg had significant contribution in this milestone. The experimental verifications of the existence of the electroweak interaction were established by the In 1973, existence of neutral currents announced by the Gargamelle collaboration and then in 1981, the discovery of two massive gauge bosons (W and Z) by UA1 and UA2 collaborations verified the existence of EWK force. These boson should be massless (like photon in QED) according to Goldstone's theorem. To overcome this issue, spontaneous symmetry breaking was introduced that generates the masses of the particles by the Higgs mechanism and separates weak interaction from electromagnetic (EM) interaction [**Reference7**, **Reference8**, **Reference9**, **Reference10**, **Reference11**, **Reference12**]. Higgs boson, a new scalar particle, was introduced by the theory being last missing patch of the SM. This triggered the search for this boson of the SM at experimental facilities which lasted for several years. One of the main design and construction motivations of the Large Hadron Collider (LHC) at the CERN was also to look for this new particle given unitary and above the Large Electron-Positron (LEP) constraints, i.e from 114 GeV to 1 TeV. After a long search, the CMS and the ATLAS collaborations of LHC, announced the existence of the heavy scalar boson (the Higgs) on 4 July 2012 [**Reference20**, **Reference21**]. Immediately after the discovery, further tests to measure its properties were begun. Mass of the newly discovered particle, the only free parameter in Higgs sector, has been measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV [**Reference22**]. Other properties such as spin, parity [**Reference21a**] and couplings of this boson to different particles [**Reference21b**, **Reference21c**] have also been measured and observed to consistent with the predictions of SM. Therefore, the new particle very probable appears as the SM Higgs boson, affiliated with EWK spontaneous symmetry breaking. SM has passed enormous experimental tests in last 50 years and no significant deviation has been seen yet. Experimental evidence of the Higgs

boson is the biggest achievement of the SM. Other success includes the experimental verification of its prior predictions about the W and Z gauge bosons, gluon, top and charm quarks.

Although SM has passed numerous experimental tests with great precision, it is not considered to be theory of everything. There are many phenomena which are not explained by SM i.e Hierarchy problem, neutrino mass and existence of dark matter and dark energy etc. There are several theoretical extensions posed in SM such as the Minimal Super Symmetrical Model (MSSM) and the Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical descriptions, such as string theory and extra dimensions. More precise Higgs boson properties measurements are needed to capture any possible deviation from SM predictions which can be a doorway to new physics.

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) in proton-proton collisions at $\sqrt{s} = 13$ TeV are presented. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of SM at the TeV energy scale. Any deviation of the integrated or the differential cross sections from SM expectations would provide a direct hint to BSM contributions. Nearly similar measurements have already been reported by the ATLAS experiment in $H \rightarrow \gamma\gamma$ [ATLAS:Hgg_8TeV, ATLAS:Hgg_8TeV_13TeV], $H \rightarrow 4\ell$ [ATLAS:H4l_8TeV, ATLAS:H4l_8TeV_13TeV] decay channels, combination of both channels [ATLAS:H4l_Hgg_8TeV, ATLAS:H4l_Hgg_8TeV_13TeV] and by the CMS experiment in $H \rightarrow \gamma\gamma$ decay mode [CMS:Hgg_8TeV].

The Higgs cross section measurement helps to probe the structure, nature of the interaction of H boson with decaying particles. Differential fiducial cross sections are measured using the full Run 2 CMS data recorded at 13 TeV corresponding to several kinematic observables which are sensitive to the H boson production mechanisms: rapidity and transverse momentum of the H boson, associated jet multiplicity and transverse momentum of leading jet. These differential distributions are sensitive to production mechanism of gluon fusion, colliding protons' PDFs, theoretical modeling of hard quarks radiations and relative contributions of different production mechanisms of H boson.

The fragmentation of the dissertation is systematized as follows. An overview of the SM of particle physics, production and decays of H boson are described in Chapter 2. Overview of the LHC and its CMS experiment are presented in Chapter 3. The calibrations and reconstructions of the physics objects within the CMS detector are described in Chapter 4. Chapter 5 presents the procedure of measurements of inclusive and differential fiducial cross sections of H boson production at $\sqrt{s} = 13$ TeV. Finally, the conclusions and outlook of the dissertation are summarized in Chapter 6.

Chapter 2

Theoretical Background

2.1 The Standard Model

The main goal of particle physics is to fully understand the nature of the constituents of the matter and the fundamental forces through which they interact. There are many theories purposed to understand the physics laws at particle level but the Standard Model(SM) is the one which stood firm against high precision tests [Reference1]. It is a collaborative theory of fundamental world, developed in 20th century which can express the fundamental particles (bosons and fermions) and the underlying interactions with the concept of mediating gauge bosons. The last unverified part of the SM, the Higgs boson, was discovered in 2012 at LHC of European Organization for Nuclear Research(CERN) [Reference20, Reference21]. Despite of incredibly successful theory of particle world, the SM does fail to fully describe the known universe.

2.2 Fundamental Particles and underlying Interactions

2.2.1 Fundamental Particles

Particle with unknown structure is called fundamental particle. Fundamental particles can be categorized into fermions (characterized by Fermi-Dirac statistics) and bosons (Bose-Einstein statistics). The spin-statistics theorem classifies the particles with integral spin as bosons, while particles with half-integral spin as fermions. Fermions includes all leptons, quarks and baryons (composite particles three quarks). Ordinary matter is composed of leptons and quarks which are categorized in three “generations” as illustrated in Table 2.1. First generation includes an “up” type quark which carries electric charge $+\frac{2}{3}$ and a “down” type quark carrying electric charge $-\frac{1}{3}$. Quarks also carry color charge (red, green, or blue) which is responsible for the strong interaction. The types of the quarks in order of increasing mass are “up”, “down”, “strange”, “charm”, “bottom” and “top”. The whole visible world is only made of light type of quarks, “up” and “down”. The heavy quarks can be produced only in high energy collisions (cosmic rays, particle accelerators) and quickly decays into “up” and “down” quarks.

Quarks interact through all three Standard Model forces, weak, strong, and electromagnetic. According to the Quantum Chromodynamics (QCD), two quarks

(quark antiquark pair) or three quarks can combine as mesons ($q_1\bar{q}_2$) or baryons ($q_1q_2q_3$) respectively to make color-neutral hadrons. Out of hadrons, protons and neutrons are most widely recognized making nuclei of matter which are formed of uud and udd quarks respectively.

Fermions	1 st Gen.	2 nd Gen.	3 rd Gen.	Charge	Interactions
Quarks	<i>up</i>	<i>charm</i>	<i>top</i>	$+\frac{2}{3}$	all
	<i>down</i>	<i>strange</i>	<i>bottom</i>	$-\frac{1}{3}$	
Leptons	<i>e</i>	<i>μ</i>	<i>τ</i>	-1	weak, electromagnetic
	<i>ν_e</i>	<i>ν_μ</i>	<i>ν_τ</i>	0	weak

TABLE 2.1: Generations, charge and interaction types of quarks and leptons.

Leptons also have three generations like quarks. Leptons only interact via electromagnetic and weak interaction, since they have no color charge. In fact, quarks and gluons are only particles carrying the color charge. Lepton generations each a charged lepton and a neutrino associated. Neutrinos are electrically neutral and carry no color charges, so interact through weak force only. Charged leptons masses are listed in order of increasing mass $m_e = 0.511$ MeV, $m_\mu = 105.7$ MeV and $m_\tau = 1.78$ GeV. Charged leptons carry -1 and their antiparticle carry $+1$ electric charge. In addition, all leptons lepton flavor quantum number: all leptons have assigned $+1$ while antileptons have assigned -1 .

Carriers of fundamental forces are bosons, more details would be described in Section 2.2.2. Electromagnetic force is mediated by photon which is massless, neutral and of spin-1. On the other hand, weak force is carried by $W^\pm$ (80.4 GeV) and Z (91.2 GeV) bosons having electric charges ± 1 and 0 respectively. At last, strong forces is carried by massless, neutral spin-1 gluons. Gluon is the only gauge boson which carries color and anti-color charge, thus it can interact with other gluons.

TABLE 2.2: Gauge bosons and their masses, coupling constants and interaction ranges for the three forces of the SM

Force	Weak	Electromagnetic	Strong
Mediator boson	$W^\pm$ and Z	photon (γ)	Gluon (g)
Mass(GeV)	80.4, 91.2 GeV	0	0
Coupling Constant	$G_F \approx 1.2 \times 10^{-5} \text{ GeV}^{-2}$	$\alpha = \frac{1}{137}$	$\alpha_s \approx 0.1$
Interaction Range	10^{-16} cm	∞	10^{-13} cm

2.2.2 Fundamental Interactions

The three fundamental interactions (electromagnetic, strong and weak) are explained three dedicated theoretical descriptions Quantum Electrodynamics (QED),

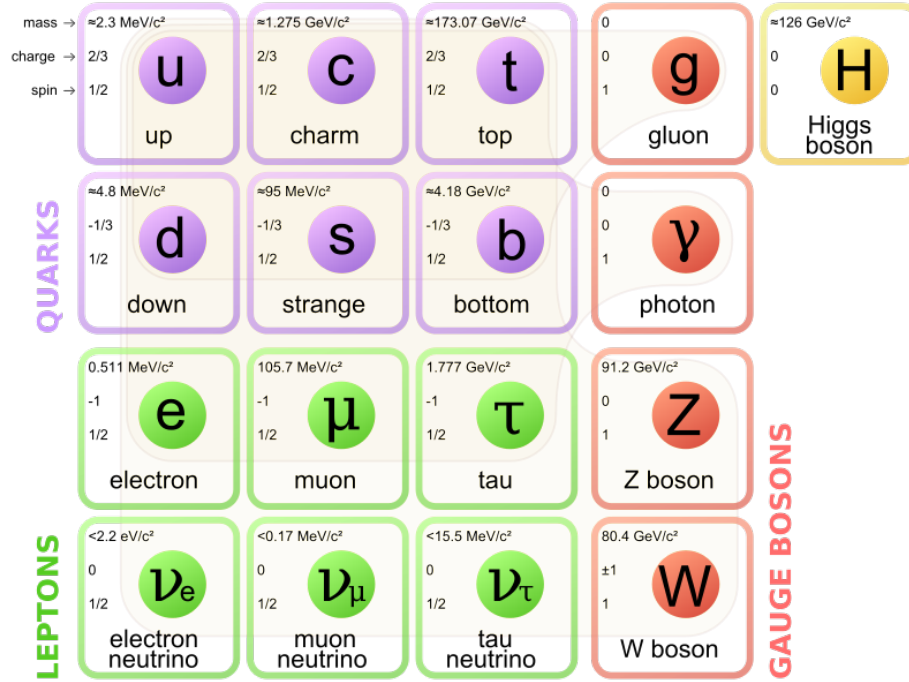


FIGURE 2.1: The elementary particles given in three generations of matter, the gauge bosons in fourth column and the Higgs boson in the fifth.

Quantum Chromodynamics (QCD) and Quantum Flavordynamics (QFD) which are encoded in SM. Electromagnetic and weak interactions were then unified into electroweak (EWK) interaction with significant contributions by Abdus Salam, Sheldon Glashow and Steven Weinberg. The mathematical description of interaction of charged particles are described by QED. Photon of QED is the force carrier of electromagnetic force. QED is the first theory which explains the interaction between light and matter that patches up special relativity and quantum mechanics. Figure 2.2 shows an example of QED, electron-positron annihilation. Annihilation is the process of destruction of particle and anti particle pair to any set of particles obeying conservation of quantum numbers, energy and momentum. QED is perturbative theory based on fundamentals of quantum mechanics established by three distinguished physicists W. Heisenberg, W. Pauli and P. A. M. Dirac during 1920s. Later on, Freeman J. Dyson, Richard P. Feynman and Julian S. Schwinger showed that charged particle can possibly interact through a sequence of processes with growing complexity, and all of these processes could be described by Feynman developed graphical technique. QED is abelian gauge theory with $U(1)$ being its symmetry group.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction [Referencea]. Weak force is mediated by $W^\pm$ and Z bosons. In terms of a field theory, it is represented by gauge theory with symmetry group $SU(2)$, having three gauge bosons (W^+ , W^- , Z). Weak force, rarely known from quantum

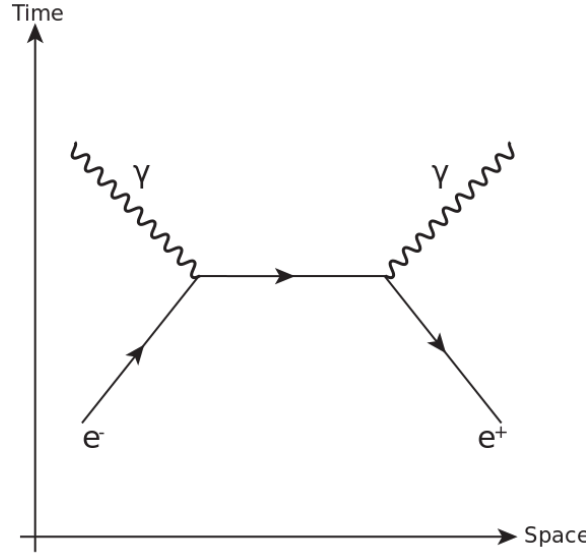


FIGURE 2.2: A Feynman diagram of annihilation of electron positron.

flavordynamics (QFD), is the only force which is proficient to change the flavor of quarks. This is very short range force because exchange bosons ($W^\pm$, Z) are massive. All fermions and the Higgs boson can interact via weak force. In 1970s, electromagnetic and weak forces were unified as electroweak force above unification energy (on the order of 100 GeV). The Physics Nobel Prize was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role in electroweak unification. In the field theory representation, Electroweak unification is settled under the gauge group $SU(2)_L \otimes U(1)_\gamma$. Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino in 1973 by Gargamelle collaboration and then the discovery of W and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [Reference2, Reference3, Reference4, Reference5].

The representation of SM is made by $SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma$ gauge symmetry. Where $U(1)_\gamma$ is representation of QED, $SU(2)_L$ represents weak theory and $SU(3)_C$ represents QCD. By the unification of electromagnetic and weak force represented by $SU(2)_L \otimes U(1)_\gamma$ gauge symmetry [Reference6], Glashow, Salam and Weinberg was able to formulate the fields to account the $W^\pm$, Z and γ (A) boson in footings of the original basis fields A' and B as expressed in equations 2.1, 2.2 and 2.3. The spontaneous breaking of $SU(2)_L \otimes U(1)_\gamma$ gauge symmetry is primary reason for the generation of masses of the particles and separate electroweak (EW) interaction from electromagnetic (EM) and weak interactions. At higher energies ($> 100 \text{ GeV}$), both weak and EM forces acts as similar style. Further details would be described in 2.3 section.

$$W_\mu^\pm = \sqrt{\frac{1}{2}}(A_\mu^1 \pm iA_\mu^2) \quad M_W = \frac{1}{2}gv \quad (2.1)$$

$$Z_\mu^\pm = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 - g'B_\mu) \quad M_Z = \frac{1}{2}v\sqrt{g^2 + g'^2} \quad (2.2)$$

$$A_\gamma = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 - g'B_\mu) \quad M_A = 0. \quad (2.3)$$

One of the the four fundamental forces is the strong force which is responsible for strong nuclear force which bounds the nucleus of an atom. As the name speaks for itself, strong force is strongest among four fundamental forces, although it act at a distance of 10^{-15} meter (femtometer) or less. Strong force is described by QCD and only exist between quarks and gluons which carry non-vanishing color charges. In the field theory representation of the SM, the strong force is represented by the symmetry group of $SU(3)_C$. Here, C represents QCD color, which must be conserved in all such interactions. The strong interaction is mediated by gluons and contrary to the other fundamental forces, strong force does not diminish with gain in distance between quarks know as color confinement. As an implication of this property, no signature of elementary quark or gluon particle are observed in High Energy Physics (HEP) colliders, instead they bound with each other to form streams of hadrons by phenomenon known as hadronization called jets.

2.3 The Brout-Englert-Higgs Mechanism (BEH)

Gauge fields must acquire mass in order to correctly describe weak interaction and to be consistent with experimental findings. Before electroweak symmetry breaking (EWSB), all particles were supposed not to have masses, however mass term in electroweak Lagrangian for a gauge boson must be introduced to guarantee the renormalization of theory that would violate gauge invariance. Which hints that some basic thing is missing in the theory apart from fermions and gauge boson fundamental fields, which can maintain the local gauge invariance. The Higgs mechanism was postulated as a part of SM, which introduces the existence of a new scalar Higgs field. This field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of masses of the fermions and bosons ($W^\pm$, Z boson) are introduced [Reference7, Reference8, Reference9, Reference10, Reference11, Reference12]. The Higgs mechanism introduces a doublet (under isospin) of complex scalar field

$$\phi = \begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^1 + i\phi^2 \\ \phi^3 + i\phi^4 \end{pmatrix} \quad (2.4)$$

which can be substituted in electroweak lagrangian given below

$$\mathcal{L}_{EWSB} = (D_\mu \phi)^\dagger (D_\mu \phi) + V(\phi^\dagger \phi) \quad (2.5)$$

here $D_\mu = \delta_\mu -igt_a W_\mu^a + \frac{i}{2}g'Y B_\mu$ represents covariant derivative and the lagrangian described in Equation 2.5 is gauge invariant under the $SU(2)_L \otimes U(1)_\gamma$ transformation. The kinetic energy term in the lagrangian is written in terms of the covariant

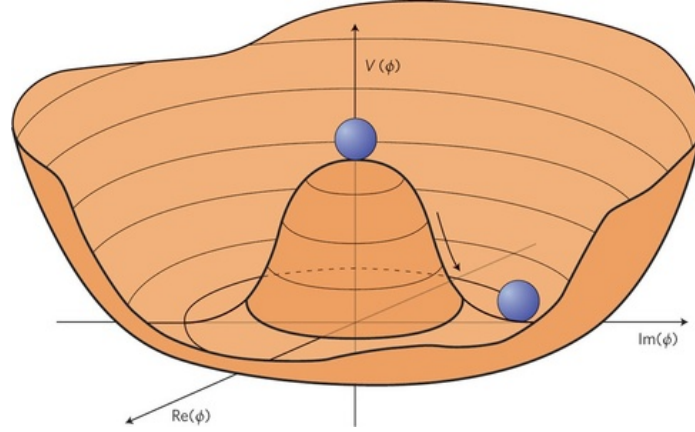


FIGURE 2.3: The Higgs potential as in Eq.2.6

derivatives whereas the potential only depends upon the product $\phi^\dagger \phi$. ϕ is the scalar field and can be codified in terms of quantum numbers which are isospin (I_3), hypercharge (Y) and electric charge (Q):

	I_3	Y	Q
$\begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix}$	$\begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

while the potential term in Equation 2.5 can also be expressed as

$$V(\phi^\dagger \phi) = -\mu^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2 \quad (2.6)$$

here μ is mass term and the parameter λ represents the strength of the interaction. In the scenario, $\mu > 0$ and $\lambda > 0$ and the potential shapes becomes very interesting, which is essential for Higgs mechanism shown in Figure 2.3. There occurs a minima for

$$\phi^\dagger \phi = \frac{1}{2}(\phi_1^\dagger + \phi_2^\dagger + \phi_3^\dagger + \phi_4^\dagger) = \frac{-\mu^2}{2\lambda} \equiv \frac{V^2}{2} \quad (2.7)$$

but on the circle of points where the magnitude of z is ϕ . This minimum does not exist at a single value of ϕ , but at circular non-zero values. The $(\phi^\dagger \phi^0)$ selection that affiliated with the ground state is arbitrary and in $(\phi^\dagger \phi^0)$ plane the selected point is not invariant under rotations which is called spontaneous symmetry breaking. To settle the ground state on the ϕ^0 axis, the vacuum expectation value (VEV) of field ϕ is

$$\langle \phi \rangle = \frac{1}{\sqrt{2}} = \begin{pmatrix} 0 \\ V \end{pmatrix}, \quad V^2 = -\frac{\mu^2}{\lambda}. \quad (2.8)$$

The field ϕ given below, is derived by rewriting it in some generic gauge, on footings of its VEV:

$$\phi = \frac{1}{\sqrt{2}} e^{i\phi^a t_a} \begin{pmatrix} 0 \\ H + \nu \end{pmatrix}, \quad a = 1, 2, 3 \quad (2.9)$$

here ϕ_a and $\phi_4 = H + \nu$ represents massless fields called Goldstone fields. New degrees of freedom is introduced along with six degrees because of polarizations vector bosons $W^\pm$ and Z , as they are massless and scalar. A unitary gauge transformation in the field is fixed by

$$\phi' = e^{-\frac{i}{\nu}\phi^a t_a} \phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ H + \nu \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi^4 \end{pmatrix}$$

so here we have zero expectation value for the Higgs field.

Now the lagrangian in Eq. 2.5 can be rewritten and disintegrated into three components

$$\mathcal{L}_{EWSB} = \mathcal{L}_H + \mathcal{L}_{HW} + \mathcal{L}_{HZ} \quad (2.10)$$

where three terms is given, assuming approximation $V = \mu^2 H^2 + \cos t$ and ignoring higher order terms:

$$\begin{aligned} \mathcal{L}_H &= \frac{1}{2} \delta_a H \delta^a H + \mu^2 H^2 \\ \mathcal{L}_{HW} &= \frac{1}{4} \nu^2 g^2 W_a W^{\dagger a} + \frac{1}{2} H W_a W^{\dagger a} \end{aligned} \quad (2.11)$$

$$\begin{aligned} &= m_W^2 W_a W^{\dagger a} + g_{HW} W_a W^{\dagger a} \\ \mathcal{L}_{HZ} &= \frac{1}{8} \nu^2 (g^2 + g'^2) Z_a Z^a + \frac{1}{4} \nu (g^2 + g'^2) H Z_a Z^a \\ &= \frac{1}{2} m_Z^2 Z_a Z^a + \frac{1}{2} g_{HZ} H Z_a Z^a. \end{aligned} \quad (2.12)$$

For $W^\pm$ and Z bosons, mass terms are included in Eqs. 2.11 and 2.12. This adds up three additional degree of freedom; however, three of the total four Goldstone bosons have also been vanished from the Lagrangian $\mathcal{L}_{EWSB}$ not affecting overall number of degrees of freedom. A new scalar field has been introduced referred as Higgs field, introduced in the theory. The boson associated of Higgs field.

Briefly, the BEH mechanism ensures the renormalizability of theory and explain existence of the massive weak gauge bosons, despite of the fact that the fundamental gauge boson in theory still stays to be massless. In fact, the symmetry still exist in equation so using “hidden” is preferred over broken. According to BEH mechanism, fermions also acquire masses via their interaction with Higgs field. The invariance of the Higgs field minimum for $U(1)_\gamma$ group shows that the electromagnetic symmetry is not broken spontaneously that keeps photon massless.

2.3.1 Masses and Couplings of Vector Bosons

Equations. 2.11 and 2.12 show that masses of $W^\pm$ and Z bosons are associated with coupling constants of electroweak and a parameter ν which is VEV that defines a

mass scale property of EWSB:

$$\begin{cases} m_W = \frac{1}{2}\nu g \\ m_Z = \frac{1}{2}\sqrt{g^2 + g'^2} \end{cases} \rightarrow \frac{m_W}{m_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos(\theta_W). \quad (2.13)$$

Couplings of Higgs boson to the bosons can also be obtained by Eqs. 2.11 and 2.12

$$g_{HW} = \frac{1}{2}\nu g^2 = \frac{2}{\nu}m_W^2 \quad (2.14)$$

$$g_{HZ} = \frac{1}{2}\nu(g^2 + g'^2) = \frac{2}{\nu}m_Z^2. \quad (2.15)$$

Using above Eqs. 2.14 and 2.14 helps to get the ratio of Higgs decay to $W^\pm$ and Z bosons

$$\frac{\mathcal{B}(H \rightarrow W^+W^-)}{\mathcal{B}(H \rightarrow ZZ)} = \left(\frac{g_{HW}}{\frac{1}{2}g_{HZ}} \right)^2 = 4 \left(\frac{m_W^2}{m_Z^2} \right)^4 \approx 2.4. \quad (2.16)$$

At last, the characteristic mass scale for EWSB may be obtained by the relation between the Fermi constant G_F and the ν parameter:

$$\nu = \left(\frac{1}{\sqrt{2}G_F} \right) \approx 246 \text{ GeV}. \quad (2.17)$$

2.3.2 Fermions Masses and Couplings

Fermions couples to Higgs field to acquire mass by Higgs mechanism. This can be achieved by introducing an invariant term $\text{SU}(2)_L \times \text{U}(1)_\gamma$ (which is called yukawa term) in SM lagrangian which can describe interactions between Higgs and the fermion fields. ϕ is an iso-doublet and the fermions can be split into right-handed and left-handed singlet. So, the yukawa term for each fermion will have expression given below for leptons:

$$\mathcal{L}_\ell = -G_\ell \bar{l}_\ell \phi l_R + \bar{\ell}_R \phi^\dagger l_\ell. \quad (2.18)$$

A mass term will only evolve for second component of l_ℓ because first part of ϕ is zero. Actually, this also depicts that the neutrinos are massless.

$$\mathcal{L}_\ell = -\frac{G_{H\ell}}{\sqrt{2}} \nu \bar{\ell} \ell - \frac{G_{H\ell}}{\sqrt{2}} H \bar{\ell} \ell. \quad (2.19)$$

For quarks, down type quarks(i.e. down, strange and bottom) behave similar to the leptons while up type quarks (i.e. up, charm, top) should couple to the charge conjugate of ϕ

$$\phi_c = -i\tau_2 \phi^* = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^3 + i\phi^4 \\ -\phi^1 + i\phi^2 \end{pmatrix} \quad (2.20)$$

in unitary gauge that becomes

$$\phi_c = \frac{1}{\sqrt{2}} \begin{pmatrix} \eta + \nu \\ 0 \end{pmatrix}. \quad (2.21)$$

Therefore, the yukawa Lagrangian can be expressed as

$$\mathcal{L} = -G_{H\ell} \bar{L}_L \phi \ell_R - G_{Hd} \bar{Q}_L \phi d_R - G_{Hu} \bar{Q}_L \phi^c u_R + h.c.. \quad (2.22)$$

The couplings constant and mass of a fermion can be derived from Eq. 2.19, written as

$$m_f = \frac{G_{Hf}}{\sqrt{2}} \nu \quad (2.23)$$

$$g_{Hf} = \sqrt{G_{Hf}} \sqrt{2} = \frac{m_f}{\nu}. \quad (2.24)$$

The fermions mass can not be predicted by the theory as G_{Hf} is free parameters. An fermion object acquires mass when Higgs field interacts with it to trans the right-handed chirality into the left-handed chirality and vice versa. The strength of the interaction of Higgs fields with particles is reflected by its mass.

This evaluation is rather interpretative, as it not accumulating the fact of three families of leptons and quarks in reality. The difference between eigenstates of weak mass and the eigenstates of physical mass makes real analysis complicated. Transformation of the weak mass basis to physical basis is essential to deal with the physical particles. For quark sector, it is achieved by Cabibbo-Kobayashi-Maskawa (CKM) matrices and in case of lepton sector, it is done by Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrices.

2.4 Higgs boson Production and its Decay

After electroweak unification, spontaneous symmetry breaking and addition of Higgs mechanism to SM solved all the puzzles with an assumption that there exists the Higgs boson. This boson was considered as the last missing subject of the SM and has been searched at LHC within full mass range allowed i.e from 114 GeV to 1 TeV (given unitary and above LEP constraints). Discussing the production and decay modes of Higgs would allow us to point out the most suitable decay channels Higgs boson discovery point of view and its properties measurements. Theory does not predict the Higgs mass; however, its couplings with bosons are proportional to the square of their masses as shown in Eqs. 2.14 and 2.15. In case of fermions, there is direct proportionality between their couplings with the Higgs and their masses as in Eq. 2.24. For this reason, the most promising channels for Higgs searches are its decays to W^+W^- , ZZ and third generation of leptons. Also, photon being massless particle does not couple to Higgs directly but through loop contributions (most probable are top quark and $W^\pm$ boson loops).

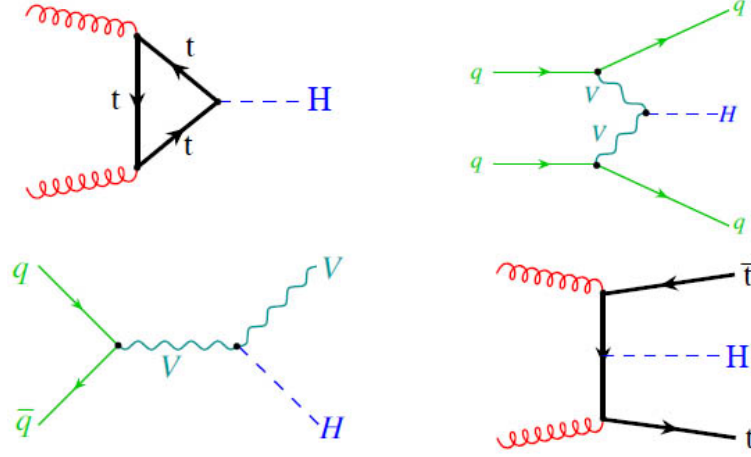


FIGURE 2.4: Feynman diagrams of Higgs production at tree level: gluon-gluon fusion (upper left); VV fusion (upper right); associated production with W and Z bosons (bottom left); associated production with $t\bar{t}$ (bottom right).

2.4.1 Higgs boson Production

The production modes of Higgs which are dominant in proton proton collisions at LHC are illustrated in Figure 2.4, corresponding production cross sections as a function of its mass at center-of-mass energies i.e. $\sqrt{s} = 7, 8$ and 13 TeV are shown in Figure 2.5. A considerable increase in production cross section of Higgs is found as we move towards higher center-of-mass energies (Figure 2.6 [Referenceb]).

Gluon-gluon Fusion

The most dominant production mode of Higgs at LHC is gluon-gluon fusion as illustrated in Figure 2.4 (top left) [Referenceb]. As LHC and tevatron are hadron colliders, it is highly probable that two gluons responsible for binding of hadron collide. The gluons couple to Higgs through the loop of virtual quarks, as coupling is directly proportional to fermion mass, top quark loop is most dominant. For SM Higgs ($m_H = 125$ GeV) the ggH cross section at 7 TeV, 8 TeV and 13 TeV are approximately 15, 19, 44 pb respectively. As we move towards higher energies the production cross section increases significantly, i.e. 27% from 7 TeV to 8 TeV and 227% from 8 TeV to 13 TeV at $m_H = 125$ GeV. The cross section numbers in Table 2.3 are evaluated at NNLO (QCD) and NLO (EW) accuracies.

Vector Boson Fusion (VBF)

VBF is the second most important production mode of Higgs at LHC and LEP as shown in Figure 2.4 (upper right). Its cross section is roughly 10 times lower than ggH at $m_H = 125$ GeV, and becomes comparable at $m_H = 1$ TeV. For SM Higgs at $m_H = 125$ GeV its cross section at 7 TeV, 8 TeV and 13 TeV are approximately 1.2,

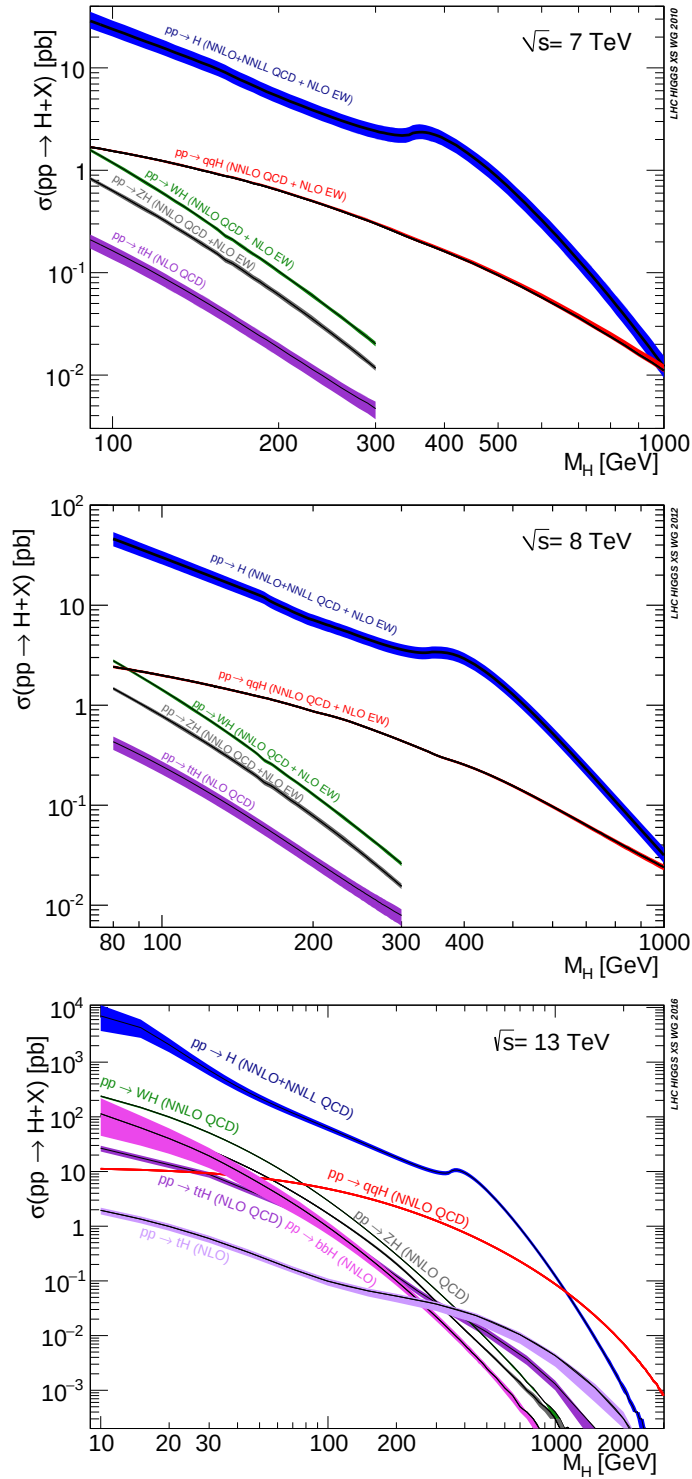


FIGURE 2.5: Production cross sections of SM Higgs boson at 7, 8 and 13 TeV as function of its mass.

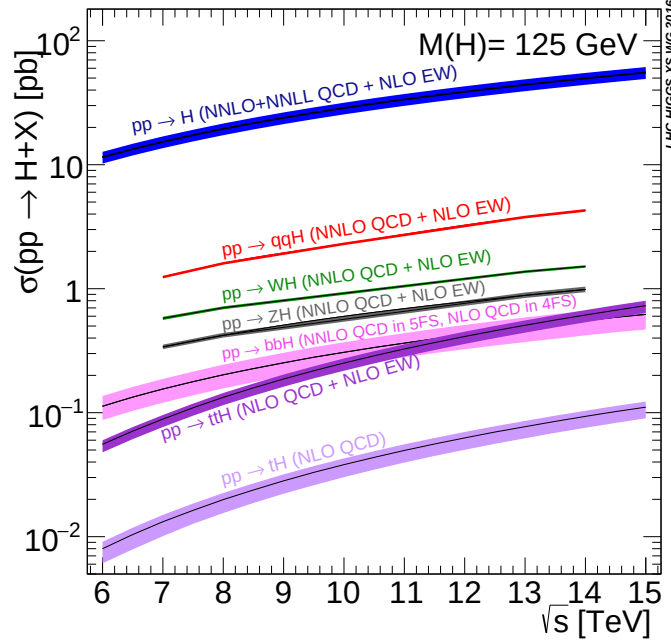


FIGURE 2.6: Production cross sections of Higgs boson ($m_H = 125$ GeV) as a function of center of mass energies

TABLE 2.3: Higgs boson production cross sections in pb for different Higgs masses and center-of-mass energies by ggH and VBF productions mechanism at LHC.

$m_H(\text{GeV})$	ggH			VBF		
	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV
120	16.61	21.07	47.38	1.30	1.68	3.9
125	15.31	19.47	44.14	1.24	1.60	3.78
130	14.15	18.03	41.23	1.19	1.53	3.64

1.6, 3.8 pb respectively. The cross section numbers in the Table 2.3 are computed at NNLL QCD and NLO EW precision [Referenceb]. In VBF production mode, two $W^\pm$ or Z boson fusion create Higgs that have been emitted by a (anti)fermion pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in detected in forward region. These jets signature helps to increase signal to background ratio.

Production of Higgs in association with W and Z bosons

Such production of Higgs is known as *HiggsStrahlung* as illustrated in Figure 2.4 (bottom left). In this process, Higgs is radiated by high energetic $W^\pm$ or Z bosons that is produced by fusion of two fermions. This production mode was leading mechanism at LEP and sub-leading mechanism at Tevatron. At LEP it becomes very

TABLE 2.4: Higgs boson production cross section in pb for different Higgs masses and center-of-mass energies by associative productions mechanism at LHC.

$m_H(\text{GeV})$	WH			ZH			ttH		
	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV
120	0.66	0.81	1.57	0.39	0.48	0.99	0.11	0.15	0.57
125	0.58	0.70	1.37	0.34	0.42	0.88	0.09	0.13	0.51
130	0.51	0.62	1.21	0.30	0.37	0.79	0.08	0.12	0.45

probable to produce $W^\pm$ or Z boson with electron positron collisions. In case of hadron collider, tevatron makes quark antiquark collisions more likely as compared to LHC. AT LHC this production mechanism is at third number. The WH and ZH cross section for Standard Model Higgs at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NNLL QCD and NLO EW precision [Referenceb].

Top Fusion

The least dominant process of Higgs production is illustrated in Figure 2.4 (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association $t\bar{t}$. The ttH cross section for Standard Model Higgs at center of mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NLO QCD and NLO EW precision [Referenceb].

2.4.2 Higgs Decay

As Higgs boson is electroweak particle, so it wants to decay quickly. Likelihood of its decay depends on many aspects, i.e. its mass, interaction strength and etc. Most of the factors are controlled by SM except the Higgs boson mass itself. The life time of SM 126 GeV Higgs is 1.6×10^{-22} s. It can decay to all SM particles with mass but it likes to decay most heavier particles available. Higgs boson decay branching ratio to a variety of particles are illustrated in Figure 2.7 [Referenceb].

$H \rightarrow b\bar{b}$ (at $m_H \approx 125$ GeV) is the most dominant decay mode of Higgs. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, CMS and ATLAS collaborations and other collaborations [Reference13], [Reference14] and [Reference15](CDF and D0) have exploited this channel, in association with a vector boson which decays to leptons, in boosted regime. B quarks are also difficult to reconstruct in a detector. When they decay in the detector, they fragment into jets of many hadrons. These jets are difficult to reconstruct perfectly. Because of this, it is necessary to exploit other channels in this region. $H \rightarrow \gamma\gamma$ is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution.

The main backgrounds in this channel are $q\bar{q} \rightarrow \gamma\gamma$, $gg \rightarrow \gamma\gamma$ and instrumental background where one or two jets are misidentified to be photons. As the Higgs mass at 125 GeV, Higgs boson decay to diboson(WW , ZZ) boson are also available,

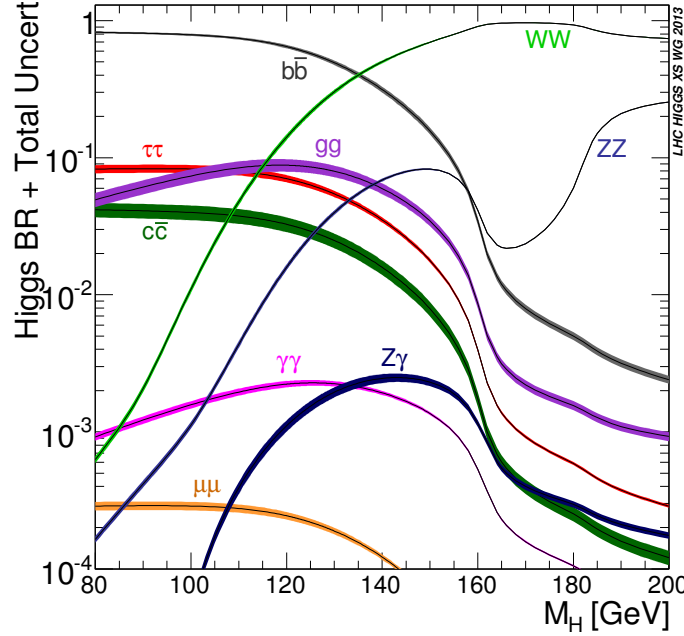


FIGURE 2.7: Branching ratios for different Higgs boson decay channels as a function of the Higgs mass.

where one or both boson can be off-shell. In case of WW decay, it's impossible to completely reconstruct WW boson in detector as it decays $H \rightarrow WW \rightarrow \ell\nu\ell\nu$ have two neutrinos and $H \rightarrow WW \rightarrow \ell\nu jj$ have one neutrino and two jets. These objects pull mass distribution to be wide and signal to background ratio to be low, which makes WW decay channel less sensitive. While $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can be reconstructed with excellence in the detector. The branching ratio of the Higgs in intermediate range is illustrated in Figure 2.8 [Referenceb].

2.5 Experimental Observations

The electroweak unification theory led to a series of impressive experimental discoveries. In 1973, the evidence of neutral current through inelastic scattering was found by Gargemelle bubble chamber collaboration at CERN [Reference16]. This evidence provided a starting platform to the Standard Model of electroweak interaction.

Super Proton Synchrotron proton-antiproton collider at CERN played its role to next breakthrough that came a decade later. The existence of W , Z boson announced by UA1 and UA2 collaboration through the channels $Z \rightarrow \mu^+\mu^-$, $Z \rightarrow e^+e^-$ and $W \rightarrow e\nu$ [Reference2, Reference3, Reference4, Reference5]. This was a

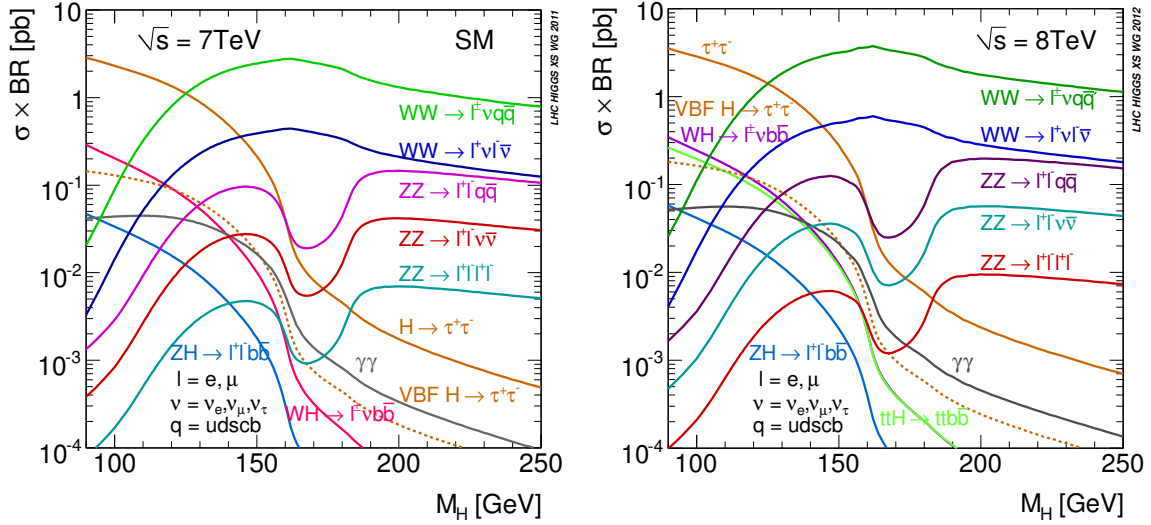


FIGURE 2.8: Cross sections of Higgs times branching ratios at 7 TeV and 8 TeV (left and right diagrams respectively).

first observation of W, Z boson that mediates weak force. For their contributions, Simon van der Meer and Carlo Rubbia were awarded with Nobel Prize for their contribution in 1984.

These important experimental discoveries and growing knowledge on weak interaction convinced to construct the LEP collider. at CERN, for deeper look into W, Z further decays, their precise mass measurement [Reference17, Reference18]. LEP was equipped with four detectors around the ring; OPAL (Omni-Purpose Apparatus for LEP), DELPHI (Detector with Lepton and Hadron Identification), ALEPH(Apparatus for LEP Physics) and L3 (Letter of Intent 3). These experiments reported many precise results of the SM, specially masses of W and Z bosons and confirming three neutrino species consistent with Standard Model [Reference18].

Around 2000, construction for the Large Hadron Collider (LHC) [Reference19] at CERN was started to look for last missing patch of SM, the Higgs boson. LHC is equipped with four major detectors; ATLAS (A Toroidal LHC Apparatus), CMS (Compact Muon Solenoid), ALICE (A Large Ion Collider Experiment), LHCb (Large Hadron Collider beauty). LHC had shown brilliant performance since being operational. It has already recorded data of proton-proton collisions at center-of-mass energies 7,8 and 13 TeV in from 2011 to 2018. After a long search , the CMS and ATLAS collaboration of LHC, certified the presence of the Higgs, a heavy scalar boson, on 4 July 2012 [Reference20, Reference21]. Further test to measure of the properties of this scalar boson immediately begins, in March 2013, CERN announced the tentative confirmation of consistency of its properties with SM [Reference22, Reference23, Reference24, Reference25].

2.6 Beyond Standard Model

Although SM has passed numerous experimental test with great precision, it is not considered to be theory of everything. There are many phenomena which are not explained by SM i.e Hierarchy problem, lack of direct detection of dark matter and dark energy, and masses of neutrinos etc. There are several theoretical extension posed in Standard Model such as Minimal Super Symmetrical Model(MSSM) and Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical description, such as string theory, extra dimension. The simplest broadening of Higgs sector of SM is the the electroweak singlet (EWS) model [Reference26]. In this model, a heavy singlet is defined along with SM scalar Higgs field and then two CP-even Higgs bosons comes out as result of spontaneous symmetry breaking. Another generic extension to SM Higgs sector is 2 Higgs Doublet Model (2HDM), which introduces two scalar, complex doublet ϕ_1 and ϕ_2 [Reference27]. Spontaneous symmetry breaking results five Higgs boson, a neutral CP-odd one (A), two charged ($H^\pm$) and two CP-even particles (h and H). Experiments are looking for any direct or indirect hints of new physics. Some of main aspects where SM is unable to explain are discussed below.

2.6.1 Hierarchy Problem

Larger discrepancy between features of the gravitational and the weak forces is known as “Hierarchy Problem”. There does exist a convincing explanation of the problem why weak force is 10^{32} times stronger than gravitational force. The field theory breaks down because of existence of quantum gravity at plank scale ($M_p \approx 10^{19}$ GeV). Writing the Standard Model Lagrangian for the Higgs mass will lead to following expression below

$$m_H^2 = m_{oH}^2 + \delta m_H^2 \quad (2.25)$$

here m_{oH}^2 represents mass parameter exist in the Standard Model Lagrangian, and Δm_H^2 represents the quantum radiative corrections to the mass that are predicted to be heavier. These corrections can be found as

$$\Delta m_H^2 \approx \frac{\lambda^2}{16\pi^2} \Lambda^2 \quad (2.26)$$

here λ represents the Yukawa coupling constant, and Λ represents the energy at which the Standard Model is invalid, or about 1 TeV. This was also one of the reason to expect Higgs boson mass at order of few hundred GeV. The Super symmetry (SUSY) provide the solution of Hierarchy problem by predicting the existence of heavier super partners of all SM particles (bosons and fermions). Several direct and indirect searches has been conducted at LHC to hunt for SUSY, but no evidence has been found so far.

2.6.2 Dark Matter and Dark Energy

Latest observation from cosmological experiments shows that universe is composed of ordinary matter, dark matter and dark energy with percentage as 4.9%, 26% and 70% respectively. SM does not provide any clue about the presence of dark matter and dark energy.

2.6.3 Matter Antimatter Asymmetry

Today, if we look around in the universe, we see matter dominance over anti-matter. Matter and anti-matter are supposed to be produced in equal quantities at the time of Big-Bang. This gives rise to one of the greatest mysteries, where all antimatter has gone. Standard Model has unable to interpret this imbalance between matter and the antimatter.

2.6.4 Grand Unification and Beyond

One of the main goals of the particle physics is to unify all fundamental force at some energy scale to describe the theory of everything as Standard Model unify electromagnetic and weak force in to electroweak. But Standard Model does not provide any clue about unification of gravity along with other fundamental force generally referred as Grand Unification Theory (GUT). This implies that Standard Model itself is a patch of Grand Unification. Search for any new physics (beyond SM) is important and justified that could improve our understanding about these unsolved mysteries.

2.7 Higgs Boson as a Probe to BSM

After the discovery of the Higgs boson, it became very natural and important to measure Higgs boson properties as precise as possible. Measurements of its properties provide a direct probe into SM Higgs sector of perturbative QCD calculations at TeV energy scale.

Couplings of Higgs boson with fermions are predicted by SM through Yukawa interactions. Through Yukawa interactions with Higgs, quarks and leptons get masses. While in SM the coupling strength of Higgs particle with other particles is directly proportional to mass of that particle. The masses of leptons and quarks are well known by experiments, so it is possible to estimate the decay rates of Higgs to fermions precisely. Experimental measurements can test these predictions which will provide a direct hint to BSM if any deviation is observed.

The cross section measurements of the Higgs boson in a process illustrate the possibility of occurrence of that process. Any deviation from SM prediction in integrated or differential cross sections will guide to BSM effects. The observation and studies of the Higgs boson production like gluon gluon fusion ($gg \rightarrow H$), vector boson fusion (VBF), production in association with a vector boson (WH, ZH) and

with a pair of top quarks ($t\bar{t}H$) also shed light on its coupling with vector boson and top quark etc.

Appendix

